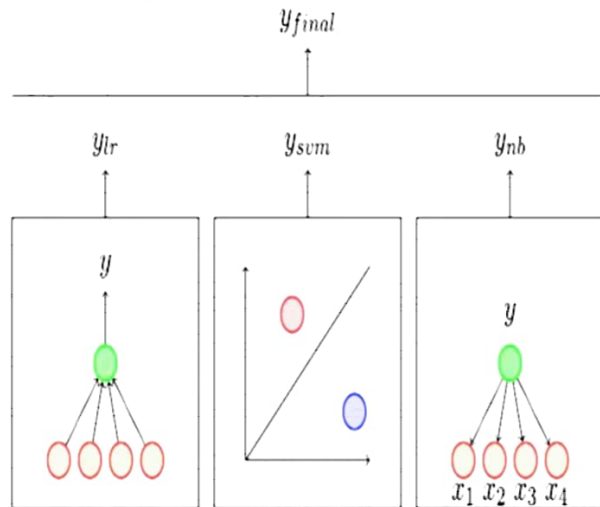


Ensemble methods

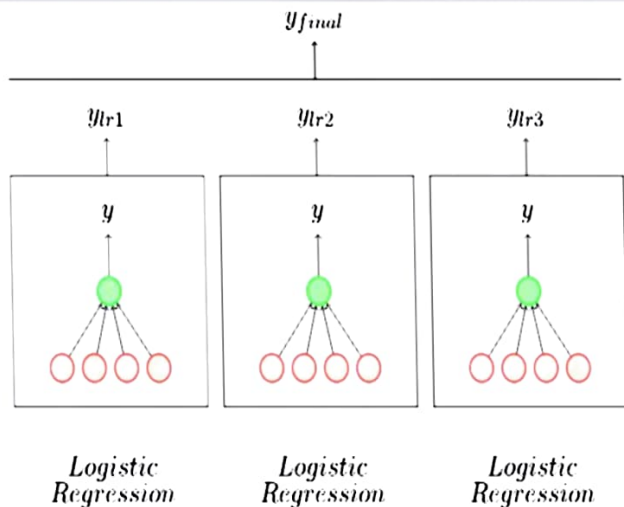


Logistic Regression

SVM

Naive Bayes

- Combine the output of different models to reduce generalization error
- The models can correspond to different classifiers
- It could be different instances of the same classifier trained with:
 - different hyperparameters
 - different features
 - different samples of the training data



Each model trained with a different sample of the data (sampling with replacement)

- Bagging: form an ensemble using different instances of the same classifier
- From a given dataset, construct multiple training sets by sampling with replacement ($T_1, T_2, \dots, T_k$)
- Train i^{th} instance of the classifier using training set T_i